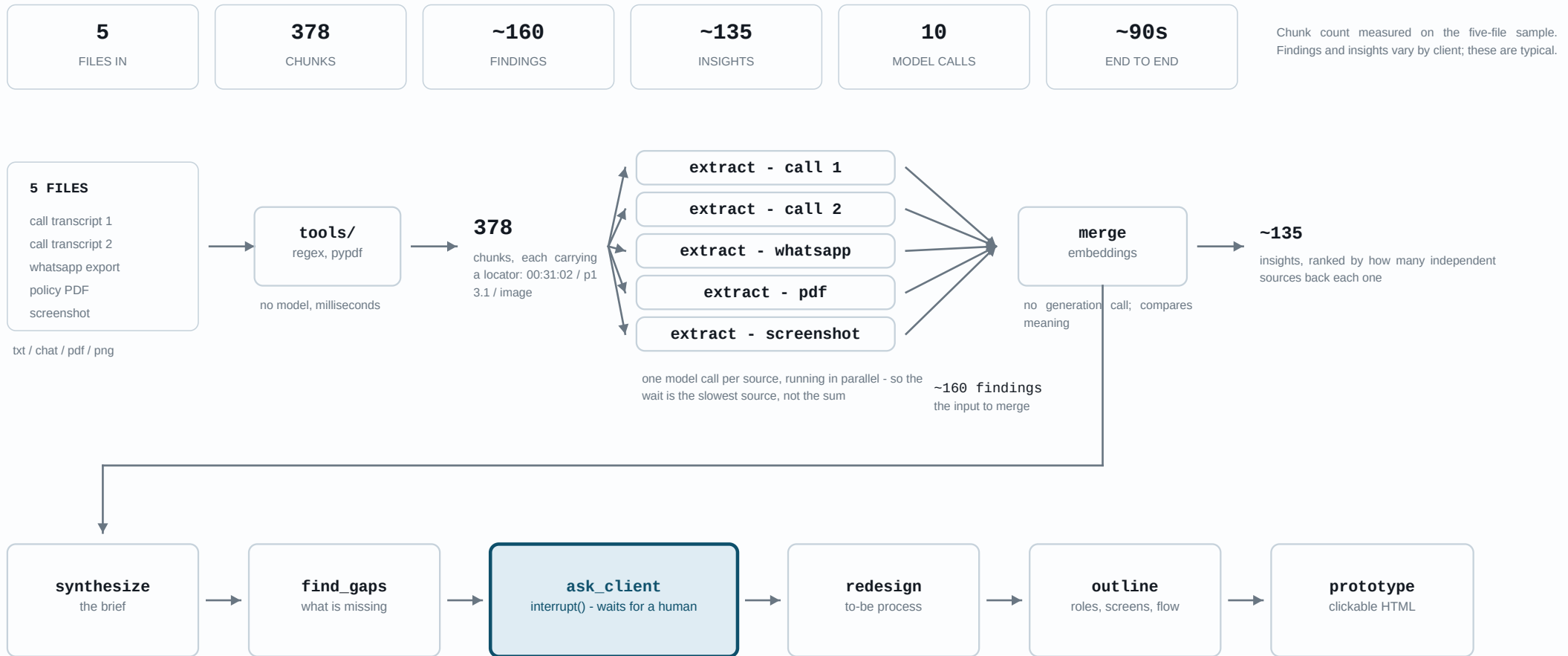


How the pipeline works

Seven agents turn scattered client material into a sourced brief, the questions nobody answered, and a prototype you can click. Parsing is ordinary code; every stage after it is a model call.



The run stops at ask_client. Nothing to the right of it is designed until a human has answered - or explicitly decided not to.

That pause is the only reason this is a graph rather than a chain of function calls. interrupt() checkpoints the entire run to disk and returns; resuming replays nothing.

What each one is for

One file per agent in backend/app/agents/, with its prompt as an editable .md beside it. Each is a plain function: state in, state updates out. No agent touches the filesystem, which is what lets the whole graph be tested with the model stubbed.

AGENT	IN -> OUT	COST	WHAT IT DOES
tools/	files -> chunks	no model	Split each format at its natural unit: an utterance, a message, a numbered clause, a paragraph. Every chunk keeps a locator, and that decides how precise a citation can be.
extractor	chunks -> findings	5 calls, low	Reads ONE source. Pulls out atomic claims tagged fact / pain / requirement / constraint / question, each citing the chunk it came from. One source at a time is what makes the source label on every finding true.
merger	findings -> insights	embeddings	Collapses the same point made in different files and counts how many independent sources back it. That count is the ranking signal for everything downstream.
synthesizer	insights -> brief	medium	Goal, current process as ordered steps, ranked pain points, requirements, constraints - and what the client ASKED for, kept deliberately separate from what the evidence says they need.
gap_finder	brief -> questions	high	Hunts three things: questions asked and never answered, sources that contradict each other, and the obvious topic nobody raised. The hardest agent and the most valuable.
ask_client	questions -> answers	PAUSE	Stops the run and hands the questions back. Answering nothing is supported - and is a different decision from never having been asked.
redesigner	brief -> to-be	medium	A better way of working, every step tagged against today: removed, automated, simplified, new, unchanged. Required to state what it does NOT fix.
outliner	to-be -> app	medium	Roles, features each naming the pain it removes, three to five screens, one end-to-end flow. Capped at five screens: a twelve-screen demo loses the client by screen three.
prototyper	app -> html	high	One self-contained page - real interactions, invented but plausible data, no backend. Validated before it is shown: truncated or network-dependent output is rejected rather than rendered.

Why you can trust any sentence it produces

An LLM handed a pile of documents will happily write a beautiful requirements document containing three requirements nobody asked for. It reads well, it is wrong, and every line sounds equally confident. So citations are not requested - they are enforced.



Invented citations are stripped and a finding left with none is discarded. A hallucinated source and a hallucinated claim are the same failure, so neither survives. Click any claim in the interface and this chain runs backwards - you land on the line a person actually said. One deliberate exception: a gap of kind 'never_discussed' may be uncited, because you cannot cite the absence of something.

Splitting is code, thinking is the model

Deciding whether a file is a WhatsApp export is a regex, not a judgment. Asking a model would cost a request, add a second, occasionally be wrong, and could never be unit-tested. tools/ never calls a model; agents/ always does.

Meaning, not word overlap

Counting shared words scored two unrelated findings at 0.33 and two identical ones at 0.29 - the wrong way round, so no threshold could work. Embeddings put the identical pair at 0.85 and the unrelated one at 0.79. Measured on real output, not assumed.

It stops before it designs

Everything after the gap list is a design decision, and making those on top of unanswered questions is how projects get rebuilt. The graph checkpoints itself to disk and waits.

The demo exists to be argued with

The prototype is throwaway on purpose: invented data, no backend, resets on refresh. Its job is to make the client point at something and say 'but we also need...' while that correction is still free.